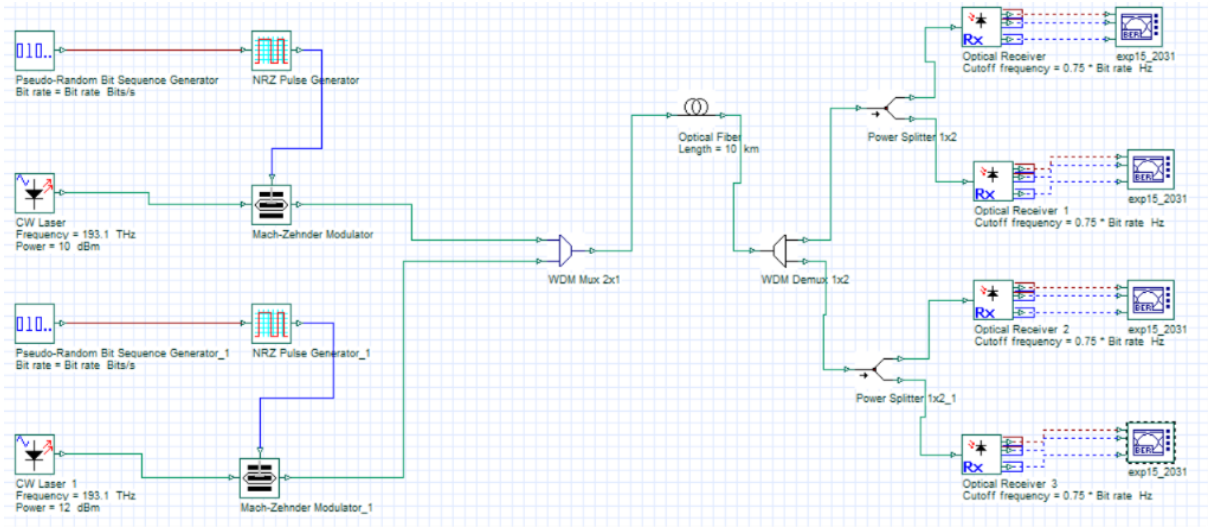
Experiment 15 : DESIGN AND PERFORMANCE ANALYSIS OF PASSIVE OPTICAL NETWORK (PON) USING OPTISYSTEM

OBJECTIVES :

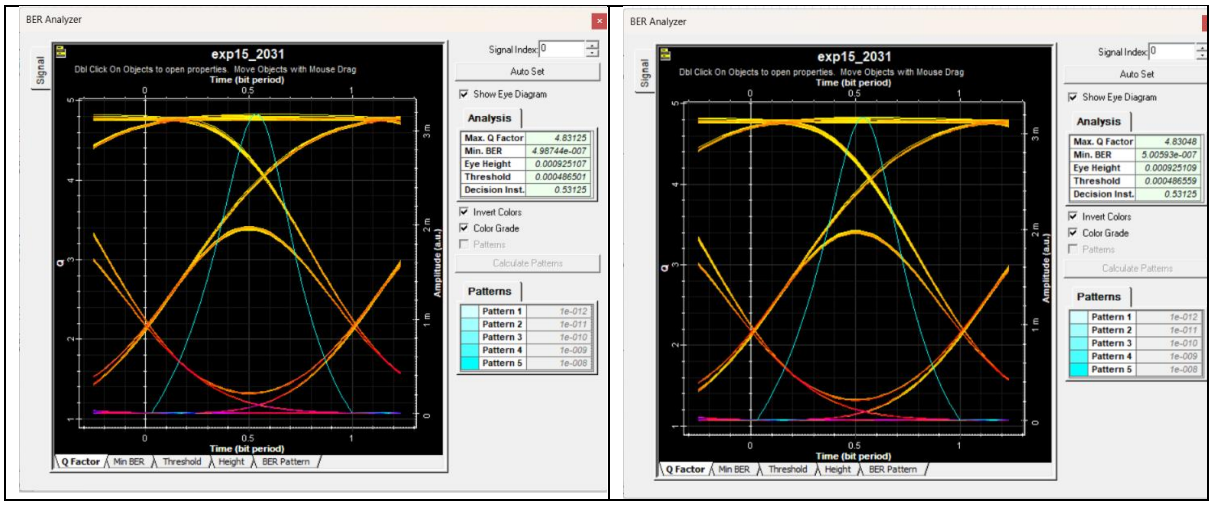
1. To design a wavelength-division multiplexed passive optical network (WDMPON) using OptiSystem.
2. To transmit multiple optical channels through a common single-mode fiber and passive optical splitter.
3. To evaluate the received signal quality using eye diagrams, Q-factor and biterror rate (BER).

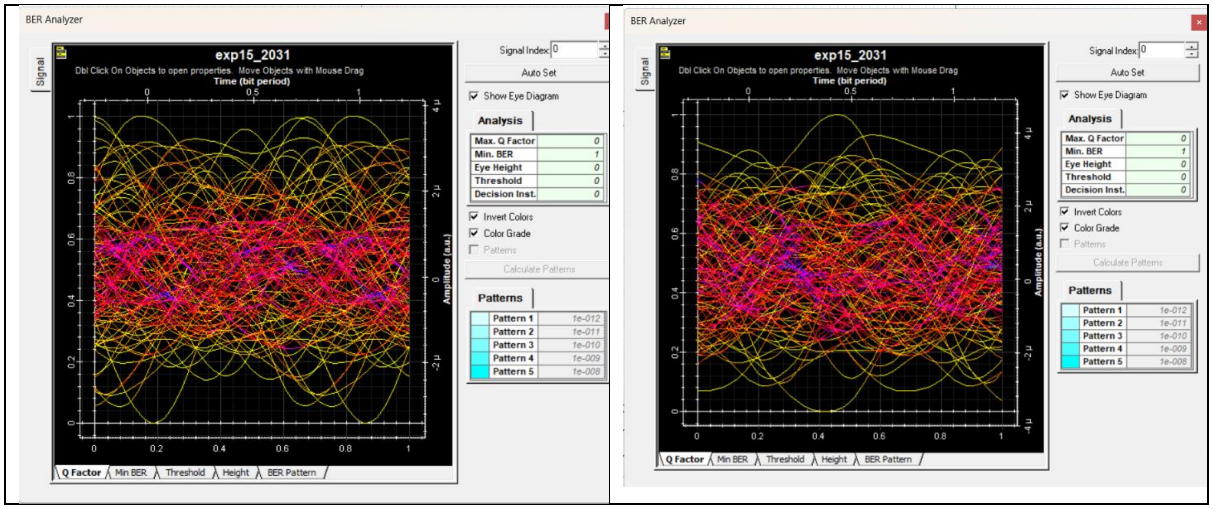
SOFTWARE USED : Optisystem

CIRCUIT DIAGRAM:



RESULTS:





REFLECTION:

I learned to design and simulate a WDM-based passive optical network (PON) using Optisystem and transmit multiple wavelength channels through a shared optical fiber and passive splitter.

I evaluated the received signal quality using eye diagrams, Q-factor and BER understanding how optical losses affect network performance

RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
System Design & WDM-PON Setup	Correct selection and connection of OptiSystem components for the WDM-PON (PRBS Generators, NRZ Generators, CW Lasers, 2×1 WDM MUX, SMF, 1×2 Passive Splitter, PIN Photodetectors with filters, BER Analyzers).	All transmitters, WDM multiplexer, fiber, splitter, receivers, and analyzers are correctly selected, connected in sequence, and configured without errors.	All components are present with minor connection or layout errors, but the network remains functional.	Component selection is partially complete; minor connection or configuration errors impede signal flow on one or more channels.	Incorrect component selection or improper setup leading to complete simulation failure.	20%
Parameter Configuration & Execution	Accurate setup of input parameters: bit rate (2.5 Gb/s), NRZ-OOK, laser wavelengths (1550.12 nm, 1550.92 nm), power (10 dBm), SMF length (20 km), attenuation (0.2 dB/km), dispersion (16.75 ps/nm/km), and splitter loss (6.02 dB).	All block parameters are accurately set and the simulation is executed for all channels.	Parameters are mostly accurate with minor setting discrepancies.	Basic parameters are configured, but multiple channel or fiber settings are missing.	Incorrect parameter settings or failure to run the simulation for the required channels.	20%

Measurement & Eye Diagram Observation	Accurate extraction and recording of Max. Q-Factor, Min. BER, Eye Height, Threshold, and Decision Instant from each BER Analyzer.	Precise data extraction for all BER Analyzers, neatly tabulated, with clear eye diagrams captured for each channel.	Data is collected with minor omissions in Q-factor, BER, or eye parameters.	Partial data collection with incomplete tables or missing eye diagrams.	Inaccurate, incomplete, or missing data and measurements.	20%
Channel Performance & Link Impairment Analysis	Detailed evaluation of per-channel performance and the effects of fiber attenuation, chromatic dispersion, and splitter loss on Qfactor, BER, and eye opening.	Comprehensive analysis correctly relating splitter loss, attenuation, and dispersion to Q-factor reduction, BER increase, and channel-to-channel variation.	Clear analysis identifying basic trends in Q-factor and BER across channels.	Basic attempt at result analysis without explaining mechanisms such as splitter loss or dispersion.	Erroneous or missing analysis of simulation results and eye diagrams.	20%
Report & Conclusion	Identification of best and worst performing channels, assessment of whether BER targets are met, and structural clarity.	Report features clear eye diagrams, complete output tables, and insightful conclusions comparing channel quality and network limits.	Satisfactory report with relevant eye diagrams and sound concluding summary.	Incomplete report with missing diagrams, tables, or weak conclusions.	Disorganized or incomplete report lacking required diagrams and valid conclusions.	20%

CONCLUSION:

A WDM-PON system was successfully designed and tested using OptiSystem. Multiple wavelengths were multiplexed, transmijed through fiber, split, and detected at the receivers. Q-factor, BER, and eye-diagram results confirmed channel performance and showed the effects of ajenuation, dispersion, and splijer loss, verifying the operation of the WDM-PON network.